University of Khartoum
Faculty of Mathematical Sciences and Informatics

GridGuard Khartoum

AI-Based Smart Load Shedding and Energy Management System for Developing Countries

Graduation project - Bachelor of Computer Science

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Application interface - 18 captured screens

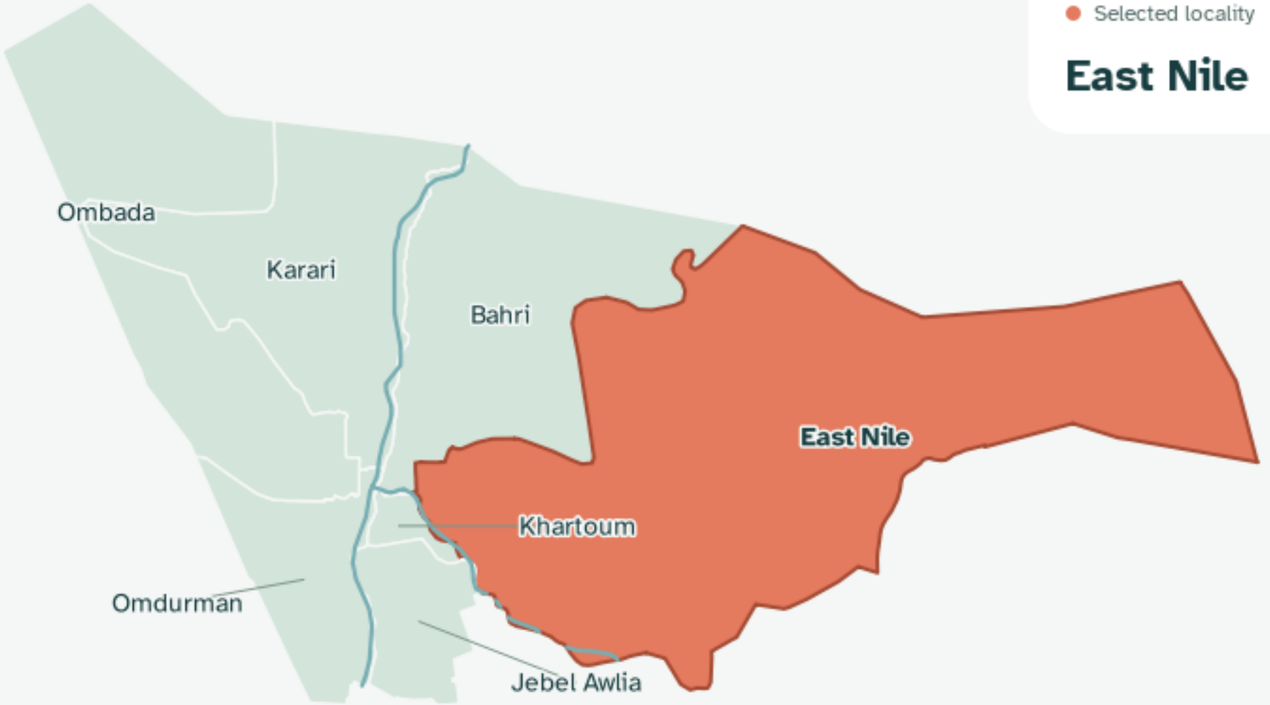
Khartoum.

Power allocation.
Seven localities in view.

Explore the current trained PPO, Fixed Rotation and constrained optimizer on one live scenario, or compare their frozen study results.

Explore scenario **Compare results**

Choose a shortage level and seed to begin.



Selected locality

East Nile

7	24	5
Localities	Hours	Service sectors

↗ PPO

Learned controller

↻ Fixed Rotation

Baseline controller

One scenario.
A matched comparison.

Synthetic scenarios · Decision support only

Boundaries: OCHA COD-AB 2020; Rivers: Natural Earth 5.0.
Geographic context, not an electrical network.

Khartoum, hour by hour.

No live scenario has been run. Choose inputs, then run a scenario.

Current trained PPO · ready to run
Synthetic scenario · Decision support only

Shortage level: Moderate

Seed: 50002

PPO | Fixed Rotation | Optimization

Run scenario

Full Service | Critical Only | Off | Selected locality



Boundaries: OCHA COD-AB 2020; Rivers: Natural Earth 5.0.
Geographic context, not an electrical network.

\+ | − | Fit

Selected locality —

East Nile

Not run

Power served	Demand
—	—

Sector service	Demand met
Hospital	—
Water	—
Bakery	—
Residential	—
Commercial	—

Next hour: unavailable · display only

▶ —

No scenario yet

00:00 | 04:00 | 08:00 | 12:00 | 16:00 | 20:00 | 23:00

Available —

Total demand —

The full 24 hours

PPO above; Fixed Rotation below.
The same scenario.

	Served energy	Critical service	Jain fairness	Supply utilization	Daily reward
	—	—	—	—	—
Fixed Rotation	—	—	—	—	—

Khartoum, hour by hour.

Explore how power reaches each locality.

PPO · seed 8111 · iteration 2500
Synthetic scenario · Decision support only

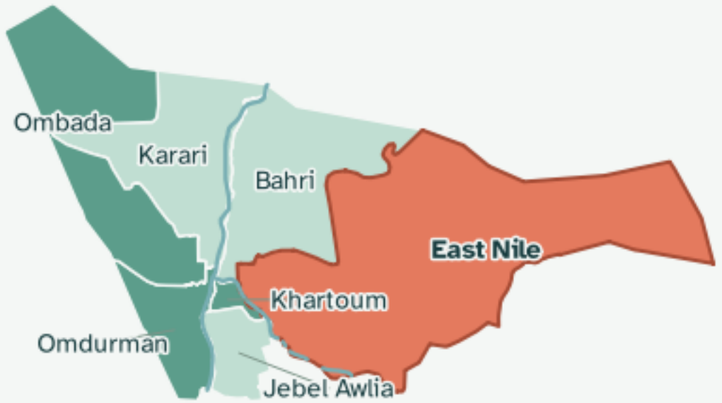
Shortage level: Moderate

Seed: 501

PPO | Fixed Rotation | Optimization

Run scenario

Full Service | Critical Only | Off | Selected locality



Boundaries: OCHA COD-AB 2020; Rivers: Natural Earth 5.0.
Geographic context, not an electrical network.

+ | − | Fit

Selected locality 12:00

East Nile

Critical Only

Power served	Demand
13.3 MW	56.6 MW

Sector service	Demand met
Hospital	100.0%
Water	100.0%
Bakery	100.0%
Residential	0.0%
Commercial	0.0%

Next-hour estimate: 56.1 MW
Display only

▶ 12:00
Hour 12 of 24

00:00 04:00 08:00 12:00 16:00 20:00 23:00

Available 271.7 MW
Total demand 429.4 MW

The full 24 hours

PPO above;
Optimization below. Same scenario.

Served energy	Critical service	Jain fairness	Supply utilization	Daily reward
6,575 MWh	100.0%	97.7%	97.7%	23.5442
Optimization 6,604 MWh	Optimization 100.0%	Optimization 95.0%	Optimization 98.2%	Optimization 23.4625

Khartoum, hour by hour.

Explore how power reaches each locality.

PPO · seed 8111 · iteration 2500
Synthetic scenario · Decision support only

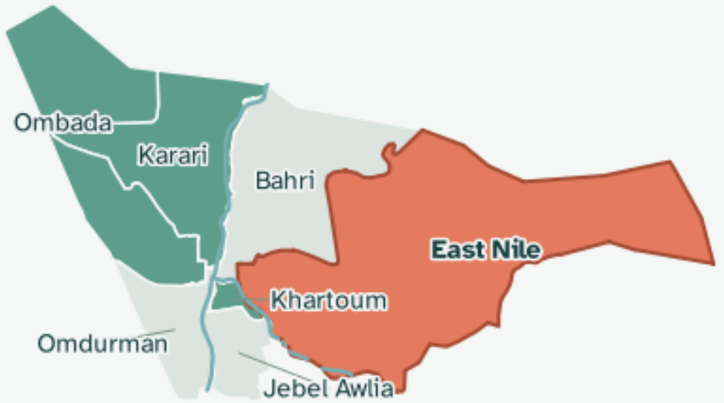
Shortage level: Moderate

Seed: 501

PPO | Fixed Rotation | Optimization

Run scenario

Full Service | Critical Only | Off | Selected locality



Boundaries: OCHA COD-AB 2020; Rivers: Natural Earth 5.0.
Geographic context, not an electrical network.

+ | − | Fit

Selected locality 12:00

East Nile

Off

Power served	Demand
0.0 MW	56.6 MW

Sector service	Demand met
Hospital	0.0%
Water	0.0%
Bakery	0.0%
Residential	0.0%
Commercial	0.0%

Next-hour estimate: 56.1 MW
Display only

▶ 12:00
Hour 12 of 24

00:00 04:00 08:00 12:00 16:00 20:00 23:00

Available 271.7 MW
Total demand 429.4 MW

The full 24 hours

Fixed Rotation above;
PPO below. Same scenario.

Served energy	Critical service	Jain fairness	Supply utilization	Daily reward
4,640 MWh	43.4%	87.8%	68.9%	13.9951
PPO 6,575 MWh	PPO 100.0%	PPO 97.7%	PPO 97.7%	PPO 23.5442

Khartoum, hour by hour.

Explore how power reaches each locality.

PPO · seed 8111 · iteration 2500
Synthetic scenario · Decision support only

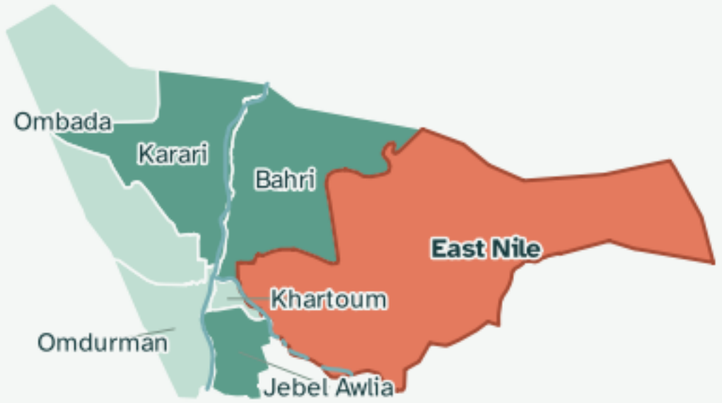
Shortage level: Moderate

Seed: 501

PPO | Fixed Rotation | Optimization

Run scenario

Full Service | Critical Only | Off | Selected locality



Boundaries: OCHA COD-AB 2020; Rivers: Natural Earth 5.0.
Geographic context, not an electrical network.

+ − Fit

Selected locality 12:00

East Nile

Full Service

Power served	Demand
56.6 MW	56.6 MW

Sector service	Demand met
Hospital	100.0%
Water	100.0%
Bakery	100.0%
Residential	100.0%
Commercial	100.0%

Next-hour estimate: 56.1 MW
Display only

▶ 12:00
Hour 12 of 24

00:00 04:00 08:00 12:00 16:00 20:00 23:00

Available 271.7 MW
Total demand 429.4 MW

The full 24 hours

Optimization above;
PPO below. Same scenario.

Served energy	Critical service	Jain fairness	Supply utilization	Daily reward
6,604 MWh	100.0%	95.0%	98.2%	23.4625
PPO 6,575 MWh	PPO 100.0%	PPO 97.7%	PPO 97.7%	PPO 23.5442

Results

Earlier results | Study details

Current study

960 matched conditions · 320 base draws · five independent PPO runs

Critical interruption

0.00 locality-h

PPO · five-run mean

Fixed Rotation 102.05 locality-h · Optimizer 0.00 locality-h

Jain service fairness

97.02%

PPO · five-run mean

Fixed Rotation 87.76% · Optimizer 94.78%

Supply utilization

97.02%

PPO · five-run mean

Fixed Rotation 71.23% · Optimizer 97.41%

Outcome	Fixed Rotation	Constrained optimizer	PPO
Critical interruption	102.05 locality-h	0.00 locality-h	0.00 locality-h
Jain service fairness	87.76%	94.78%	97.02%
Supply utilization	71.23%	97.41%	97.02%
Daily reward	13.9688	23.4468	23.4967
Switching rate	25.00%	20.55%	27.76%

Against the constrained optimizer: fairness +2.237 pp · utilization −0.399 pp · daily reward +0.0499.

Current study

960 matched conditions · 320 base draws · five independent PPO runs

PPO vs Fixed Rotation | PPO vs optimizer | Individual runs | Shortage levels | Evidence

Paired 97.5% intervals adjusted for the two primary comparisons. pp = percentage points; PPO means weight all five runs equally.

Outcome	PPO difference	97.5% interval
Critical interruption	−102.0500	[−102.0812, −102.0250] locality-h
Jain service fairness	+9.2600	[+9.1423, +9.3712] pp
Supply utilization	+25.7815	[+25.7017, +25.8594] pp
Daily reward	+9.5279	[+9.5184, +9.5357]
Switching rate	+2.7639	[+2.0601, +3.6639] pp

Fixed Rotation uses Off/Full Service without critical reservation. This compares complete systems; it does not isolate learning alone.

Close

Current study

960 matched conditions · 320 base draws · five independent PPO runs

PPO vs Fixed Rotation | PPO vs optimizer | Individual runs | Shortage levels | Evidence

Paired 97.5% intervals adjusted for the two primary comparisons. pp = percentage points; PPO means weight all five runs equally.

Outcome	PPO difference	97.5% interval
Critical interruption	+0.0000	[+0.0000, +0.0000] locality-h
Jain service fairness	+2.2368	[+2.0955, +2.3731] pp
Supply utilization	−0.3989	[−0.4642, −0.3408] pp
Daily reward	+0.0499	[+0.0406, +0.0577]
Switching rate	+7.2183	[+6.5310, +8.1124] pp

Both protect critical demand. The optimizer maximizes the current hour; the result does not compare PPO with a full-day optimum.

Close

Current study

960 matched conditions · 320 base draws · five independent PPO runs

PPO vs Fixed Rotation | PPO vs optimizer | Individual runs | Shortage levels | Evidence

Frozen final-test means on matched scenarios.

PPO seed	Selected iteration	Fairness	Utilization	Reward
1729	2900	96.89%	97.08%	23.4835
2718	3000	96.87%	97.07%	23.4913
3141	2900	97.12%	97.01%	23.5034
5772	3000	97.08%	96.99%	23.5014
8111	2500	97.14%	96.92%	23.5040

Close

Current study

960 matched conditions · 320 base draws · five independent PPO runs

PPO vs Fixed Rotation | PPO vs optimizer | Individual runs | Shortage levels | Evidence

Frozen final-test means on matched scenarios.

Shortage	Controller	Fairness	Utilization	Reward
Mild	Fixed Rotation	87.74%	54.28%	13.1911
Mild	Constrained optimizer	96.41%	98.09%	23.5881
Mild	PPO	97.99%	98.12%	23.6098
Moderate	Fixed Rotation	87.74%	68.69%	13.9600
Moderate	Constrained optimizer	95.00%	98.02%	23.4626
Moderate	PPO	97.21%	97.52%	23.5002
Severe	Fixed Rotation	87.80%	90.73%	14.7554
Severe	Constrained optimizer	92.94%	96.13%	23.2897
Severe	PPO	95.86%	95.40%	23.3801

Close

Current study

960 matched conditions · 320 base draws · five independent PPO runs

PPO vs Fixed Rotation | PPO vs optimizer | Individual runs | Shortage levels | Evidence

Pinned evidence and independent replay

3,000 iterations per run · 92.16 million training transitions total

PPO starts from fairness-rule imitation, then trains with critical-service constraints.

Zero critical interruptions and power-balance violations for PPO and the constrained optimizer.

Study statistics were verified on CUDA. Independent replay checked every input hash, 2,304 final trajectories and 5,280 validation trajectories with zero discrepancy.

data/evidence/focused-ppo-study.json

SHA-256 · a4d5a849612f06eedef96e0d28e08247e5e3757eba0b4f2e4f1a893e9d4b8247

data/evidence/focused-ppo-training.json

SHA-256 · e4f8b8f7c73693f713ecef69a06434ce75a3a101d2541654d7009398c4eb5624

docs/focused-ppo-validation.json

SHA-256 · df1cc59a3a1cd94f8333f77693dc47ef1afb6007093d940dd9deab06383a1fe1

Live Scenario uses one validation-selected PPO policy from this training study with the same critical-service constraints and constrained optimizer. A single scenario can differ from these frozen five-run benchmark means.

Close

Earlier frozen policy · separate study and scenarios

Results

Frozen holdout results

PPO and Fixed Rotation on identical matched scenarios.

Evidence details | Extended study

960 matched scenario conditions from 320 independent base draws

Mild, Moderate and Severe shortage

Critical-outage locality-hours

Lower is better

PPO **20.2**

Fixed Rotation 102.1

0 — 168 locality-hours per episode

Jain fairness

Higher is better

PPO **96.91%**

Fixed Rotation 87.76%

0% — 100%

Utilization

Higher is better

PPO **97.61%**

Fixed Rotation 71.21%

0% — 100%

The paired difference

Paired 95% intervals · pp = percentage points

Metric	PPO	Fixed Rotation	PPO – Fixed Rotation
Critical-outage locality-hours	20.2 locality-hours	102.1 locality-hours	-81.9 [-82.5, -81.2] locality-hours
Jain fairness	96.91%	87.76%	+9.15 [9.11, 9.19] pp
Utilization	97.61%	71.21%	+26.39 [26.34, 26.45] pp

Differences use unrounded values.

Power balance: maximum power-balance violation rate 0.00% across both controllers.

Comparison scope. Fixed Rotation is supply-blind, uses only Off and Full Service, and cannot select Critical Only.

Synthetic scenarios · Decision support only.
Frozen results stay separate from the live scenario.

Close

Evidence details

Close

Frozen documented-seed-disjoint holdout. Live demonstrations cannot replace these results.

Evidence artifact

gridguard-robustness-supplement-v2

Release

gridguard-khartoum-v1.2.0

SHA-256

6b56d149331ee191302cd9e6f22365988db0ae301f9441190b61214acf832e48

Hash verified · Frozen

320 documented-seed-disjoint independent base draws × 3 shortage levels = 960 matched scenario conditions · 1,920 controller evaluations

Interpreting the evidence

Base paired 95% confidence intervals appear in the Results table. Sensitivity: pointwise 95%; no multiplicity adjustment or result gate. There is no favorable/adverse release gate.

Seeds are disjoint only from every documented training, validation, and model-selection seed in data/model/training-provenance.yaml. The holdout was not preregistered or independently custodied and cannot rule out undocumented activity.

Comparator boundary: Fixed Rotation is supply-blind, limited to Off and Full Service, and cannot select Critical Only. Synthetic decision support only, not field evidence. No behavior-cloning-only ablation was run, so this comparison cannot isolate the incremental effect of PPO fine-tuning.

Sensitivity intervals are descriptive, pointwise 95%, and unadjusted for multiple comparisons; they do not support family-wide significance claims.

The power-balance result is a simulator conservation invariant, not evidence of real-grid operational safety. The accepted reproduction is historical parity evidence; its three seeds participated in model selection.

Study limitations

Synthetic demand traces do not establish real-world forecast accuracy.

Sensitivity covers only one-factor-at-a-time ±25% demand-allocation perturbations.

Sensitivity intervals are descriptive, pointwise 95%, and unadjusted for multiple comparisons; they do not support family-wide significance claims.

Seeds are disjoint only from every documented training, validation, and model-selection seed in data/model/training-provenance.yaml. The holdout was not preregistered or independently custodied and cannot rule out undocumented activity.

No behavior-cloning-only ablation was run, so this comparison cannot isolate the incremental effect of PPO fine-tuning.

Extended study

960 matched scenario conditions from 320 independent base draws; 2,880 controller evaluations.

Mild, Moderate and Severe shortage. Evaluation seeds: 91001, 91002, 91003, 91004, 91005, 91006, 91007, 91008, 91009, 91010. Batch size: 32.

Controllers | Paired differences | Forecasting

Metric	Fixed Rotation	PPO	Optimization
Critical interruption (locality-hours)	102.07	19.86	6.75
Cumulative-service Jain fairness	87.76%	96.89%	94.89%
Supply utilization	71.24%	97.62%	98.24%
Critical energy served	42.63%	99.70%	99.87%
Zero-service blackout (locality-hours)	96.00	0.00	0.00
Blackout-duration disparity (hours)	1.71	0.00	0.00
Critical-duration disparity (hours)	1.90	0.00	0.00
Longest blackout per episode (hours)	6.00	0.00	0.00

Optimization exhaustively maximizes the canonical one-hour reward over 2,187 service-tier actions. It uses current and past state; it is not a 24-hour optimum.

Blackout means zero service when demand is positive. Critical interruption means less than 99% of critical demand served. Duration disparity is the mean gap across the 21 locality pairs within each episode; equal durations can still mean poor service. Global proportional clipping can give every locality the same critical interruption hours.

Synthetic scenarios; decision support only.
Evidence: gridguard-proposal-supplement-v1

Close

Extended study

960 matched scenario conditions from 320 independent base draws; 2,880 controller evaluations.

Mild, Moderate and Severe shortage. Evaluation seeds: 91001, 91002, 91003, 91004, 91005, 91006, 91007, 91008, 91009, 91010. Batch size: 32.

Controllers | **Paired differences** | Forecasting

Compare: PPO - Fixed Rotation

Metric	Mean difference	Paired 95% interval
Critical interruption (locality-hours)	-82.22 locality-h	[-82.91, -81.53] locality-h
Cumulative-service Jain fairness	+9.13 pp	[+9.10, +9.17] pp
Supply utilization	+26.38 pp	[+26.33, +26.43] pp
Blackout-duration disparity (hours)	-1.71 h	[-1.71, -1.71] h
Critical-duration disparity (hours)	-1.90 h	[-1.90, -1.90] h

Difference = first controller minus second. Lower interruption durations and duration gaps are better; higher Jain fairness and utilization are better. pp means percentage points. Intervals are descriptive and pointwise.

Synthetic scenarios; decision support only.
Evidence: gridguard-proposal-supplement-v1

Close

Extended study

960 matched scenario conditions from 320 independent base draws; 2,880 controller evaluations.

Mild, Moderate and Severe shortage. Evaluation seeds: 91001, 91002, 91003, 91004, 91005, 91006, 91007, 91008, 91009, 91010. Batch size: 32.

Controllers | **Paired differences** | Forecasting

Compare: Optimization - Fixed Rotation

Metric	Mean difference	Paired 95% interval
Critical interruption (locality-hours)	-95.32 locality-h	[-95.72, -94.91] locality-h
Cumulative-service Jain fairness	+7.13 pp	[+7.05, +7.21] pp
Supply utilization	+27.00 pp	[+26.95, +27.05] pp
Blackout-duration disparity (hours)	-1.71 h	[-1.71, -1.71] h
Critical-duration disparity (hours)	-1.90 h	[-1.90, -1.90] h

Difference = first controller minus second. Lower interruption durations and duration gaps are better; higher Jain fairness and utilization are better. pp means percentage points. Intervals are descriptive and pointwise.

Synthetic scenarios; decision support only.
Evidence: gridguard-proposal-supplement-v1

Close

Extended study

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Controllers | **Paired differences** | Forecasting

Compare: PPO - Optimization

Metric	Mean difference	Paired 95% interval
Critical interruption (locality-hours)	+13.10 locality-h	[+12.32, +13.86] locality-h
Cumulative-service Jain fairness	+2.00 pp	[+1.93, +2.08] pp
Supply utilization	-0.62 pp	[-0.65, -0.59] pp
Blackout-duration disparity (hours)	+0.00 h	[+0.00, +0.00] h
Critical-duration disparity (hours)	+0.00 h	[+0.00, +0.00] h

Difference = first controller minus second. Lower interruption durations and duration gaps are better; higher Jain fairness and utilization are better. pp means percentage points. Intervals are descriptive and pointwise.

Synthetic scenarios; decision support only.
Evidence: gridguard-proposal-supplement-v1

Close

Extended study

960 matched scenario conditions from 320 independent base draws; 2,880 controller evaluations.

Mild, Moderate and Severe shortage. Evaluation seeds: 91001, 91002, 91003, 91004, 91005, 91006, 91007, 91008, 91009, 91010. Batch size: 32.

Controllers | Paired differences | Forecasting

One-hour-ahead demand prediction

Learned forecast fitted on 768 training traces from seeds 42, 43, 44. 320 evaluation traces use the separate study seeds above, targeting hours 3–23.

Forecast method	Locality-sector MAE (MW)	Locality-sector RMSE (MW)	System total MAE (MW)	System total RMSE (MW)
Trailing mean (3 hours)	2.3385	3.7436	42.7512	49.9759
Persistence (1 hour)	1.3465	2.1776	24.3809	28.5917
Learned ridge AR(3)	0.6573	1.1882	9.3376	11.9242

MAE is mean absolute error; RMSE is root mean square error. Lower is better. Locality-sector errors and total-system errors use different aggregation scales. This forecast comparison does not retrain or alter the PPO policy.

Synthetic scenarios; decision support only.
Evidence: gridguard-proposal-supplement-v1

Close